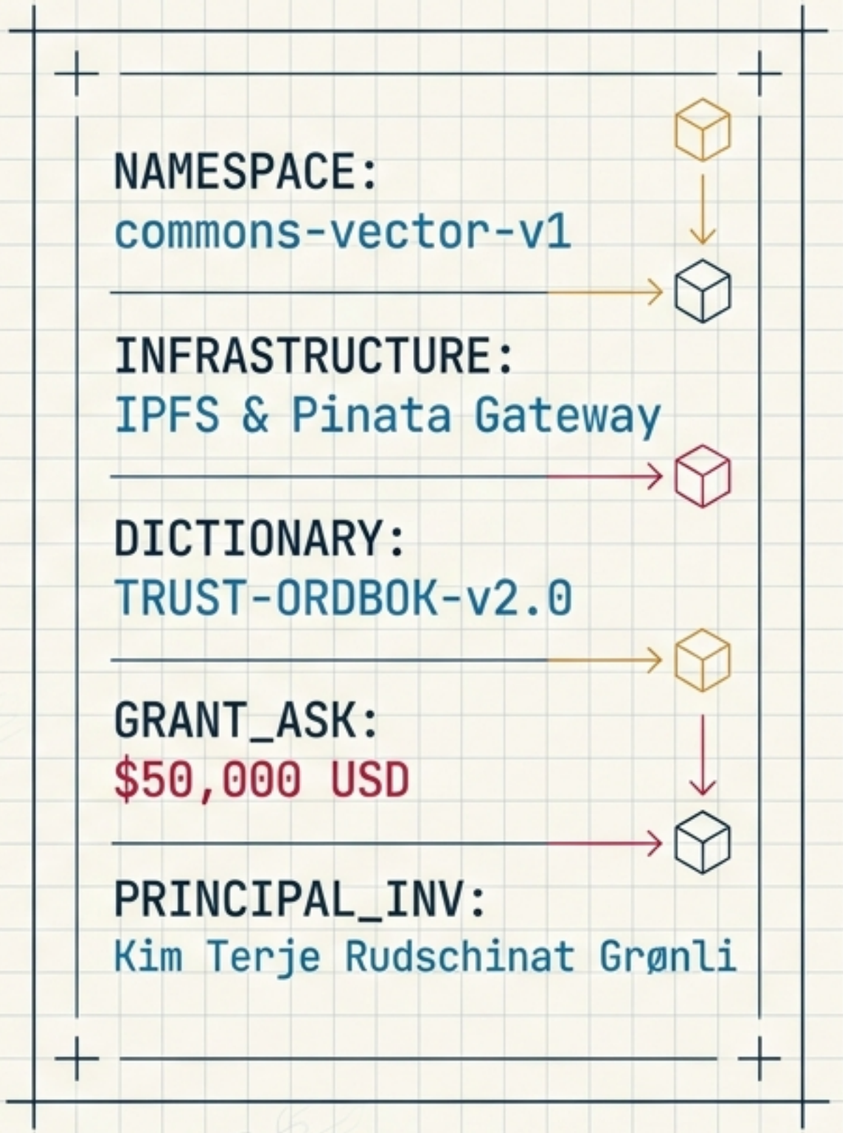
Commons Vector Protocol (CVP) & Det Levende Arkivet

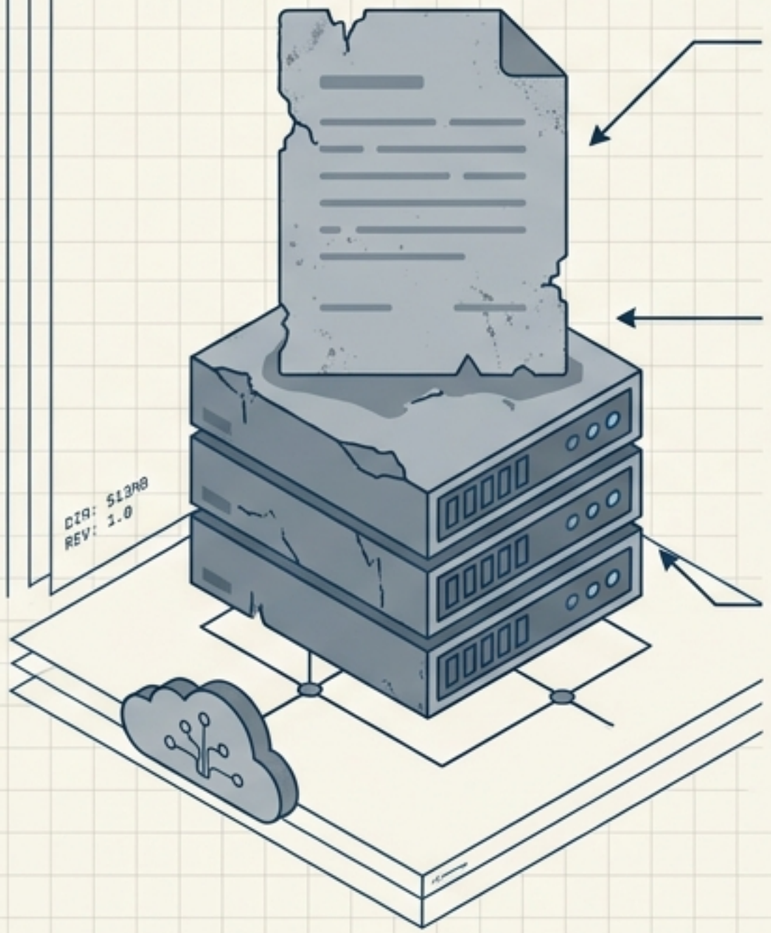
Transforming IPFS into a **self-verifying**, immutable historical record.



Gemini Notebook

The Provenance Void in Data Storage

Legacy Centralized Registries



Absence of Memory: No inherent record of creation or modification lineage.

Tamper-Prone: Administrators can alter, backdate, or delete without audit trails.

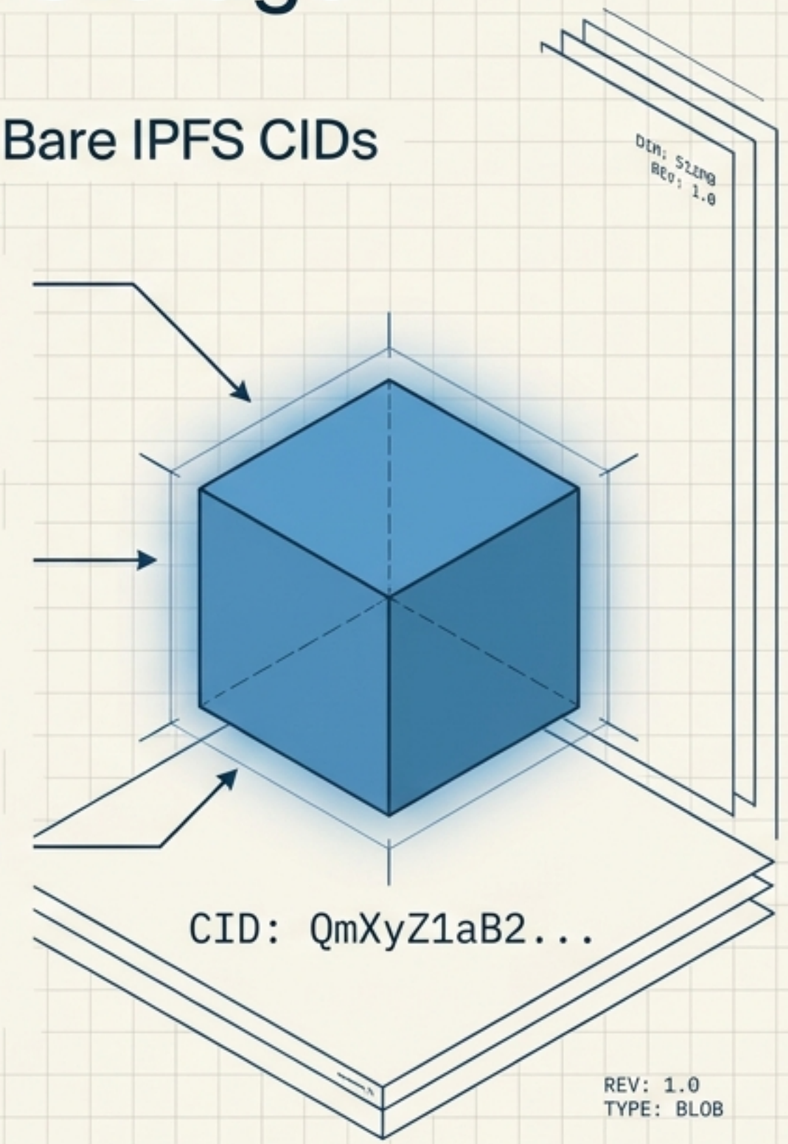
Conflicting Claims: Permits duplicate issuance for the same asset.

Bare IPFS CIDs

Static Content Addressing: Identifies what data is, but suffers from contextual drift.

Lacks Spatial Memory: Fails to encode where it resides in a namespace.

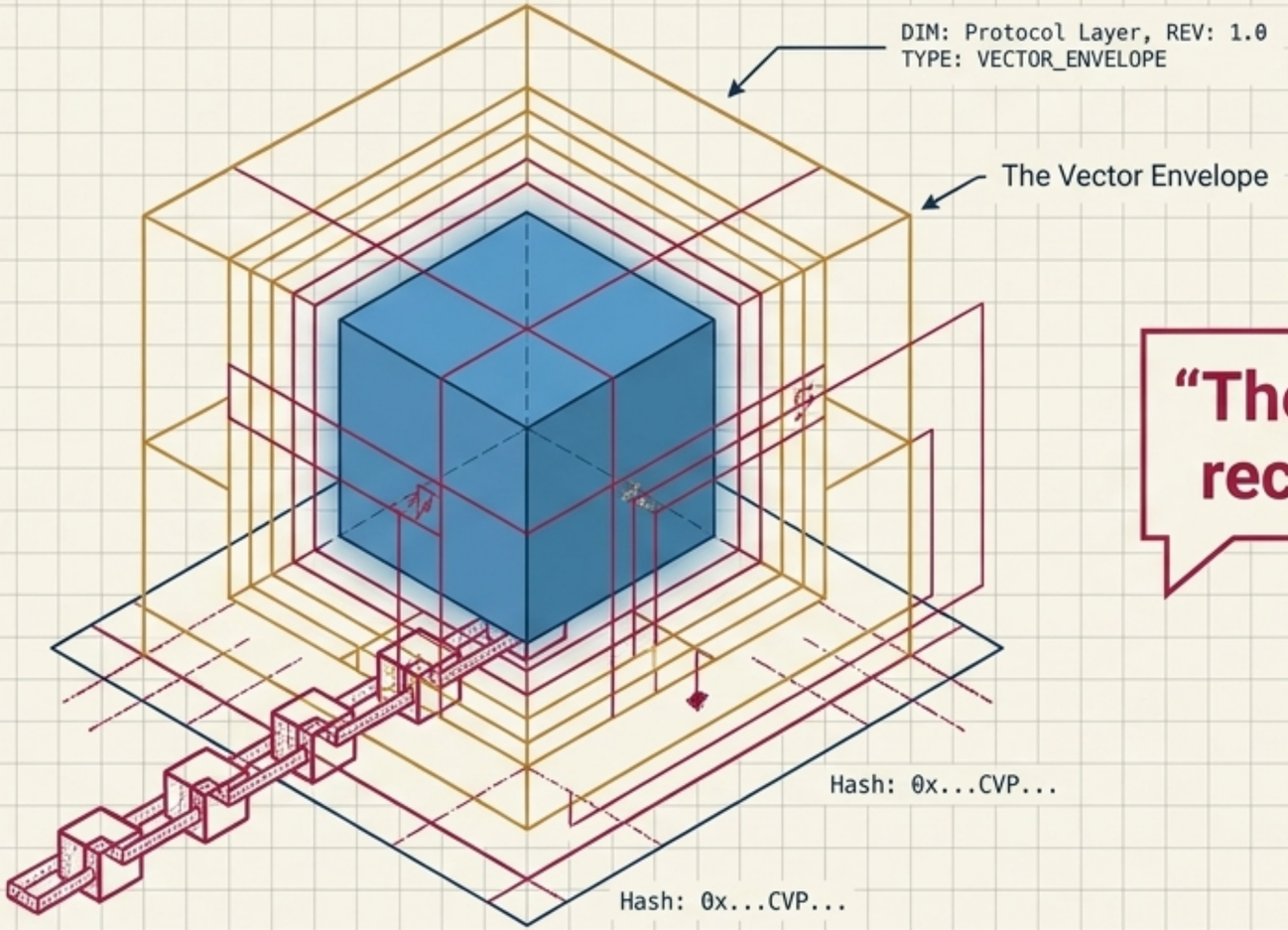
Lacks Governance: Omits how it is governed or who signed it in sequence.



Gemini Notebook

The Commons Vector Protocol (CVP)

CVP unifies static IPFS content addressing with multi-dimensional contextual vectors, Ed25519 signatures, and sequence chains. It establishes a gasless, lightweight provenance framework natively on IPFS primitives.



“The first record wins.”

- Active, self-verifying historical record.
- Operates independent of central authorities.
- Priority and authenticity through direct mathematical proof.

Gemini Notebook

The Tri-Vector Architecture (File-ID3)

Content Vector (cid)

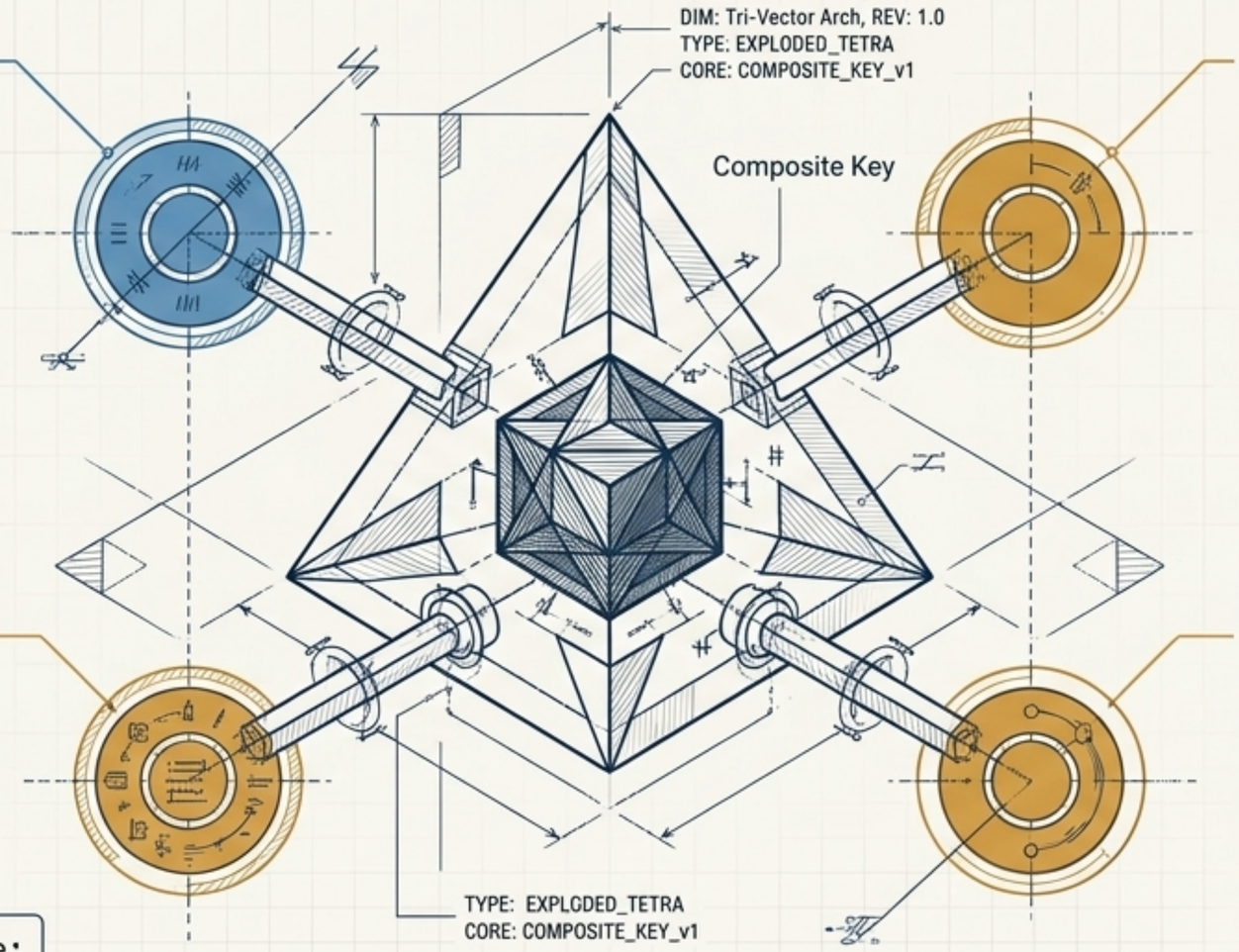
What it is:
Cryptographic IPFS hash of raw payload.

```
bafkreidkpyd...
```

Spatial Vector (v1_spatial)

Where it resides:
Namespace address within the trust hierarchy.

```
trust:commons-vector-v1:living:witness
```



Rule Vector (v2_rule)

How it is governed:
Governance schema and lexical dictionary.

```
ILPJ-v2.0 / format:native; extension:.car
```

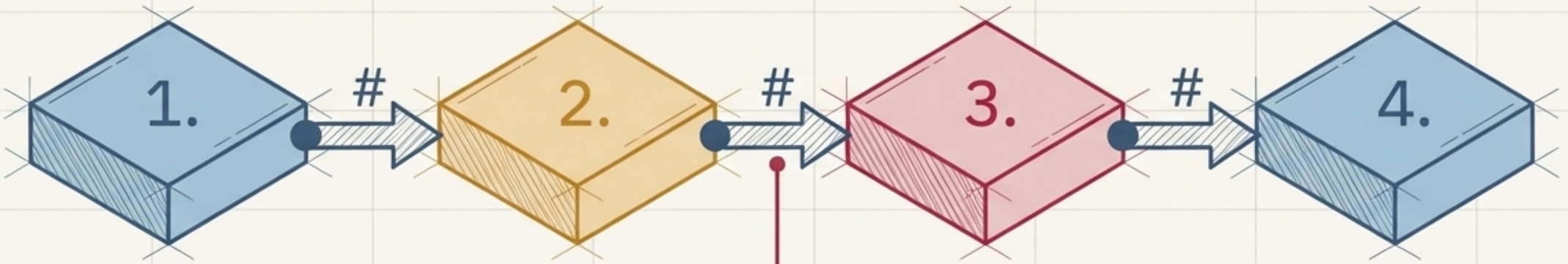
Relation Vector (v3_relation)

How it connects:
Linkage block defining explicit relationships.

```
type: derived_from;
target: bafkreigq3...
```

Gemini Notebook

Cryptographic Core & Provenance Chaining



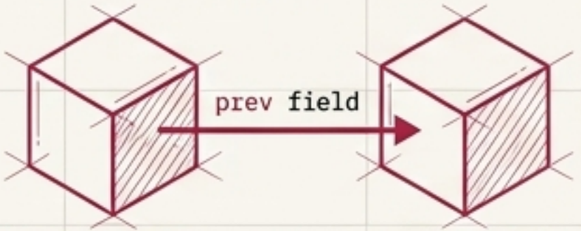
`Composite Key = SHA512(CID : Spatial : Rule)`

1. Canonical Serialization
Applies JSON Canonicalization Scheme (JCS, RFC 8785) to eliminate formatting variances.

2. Composite Key Derivation
Locks payload, location, and rules into one mathematical truth.

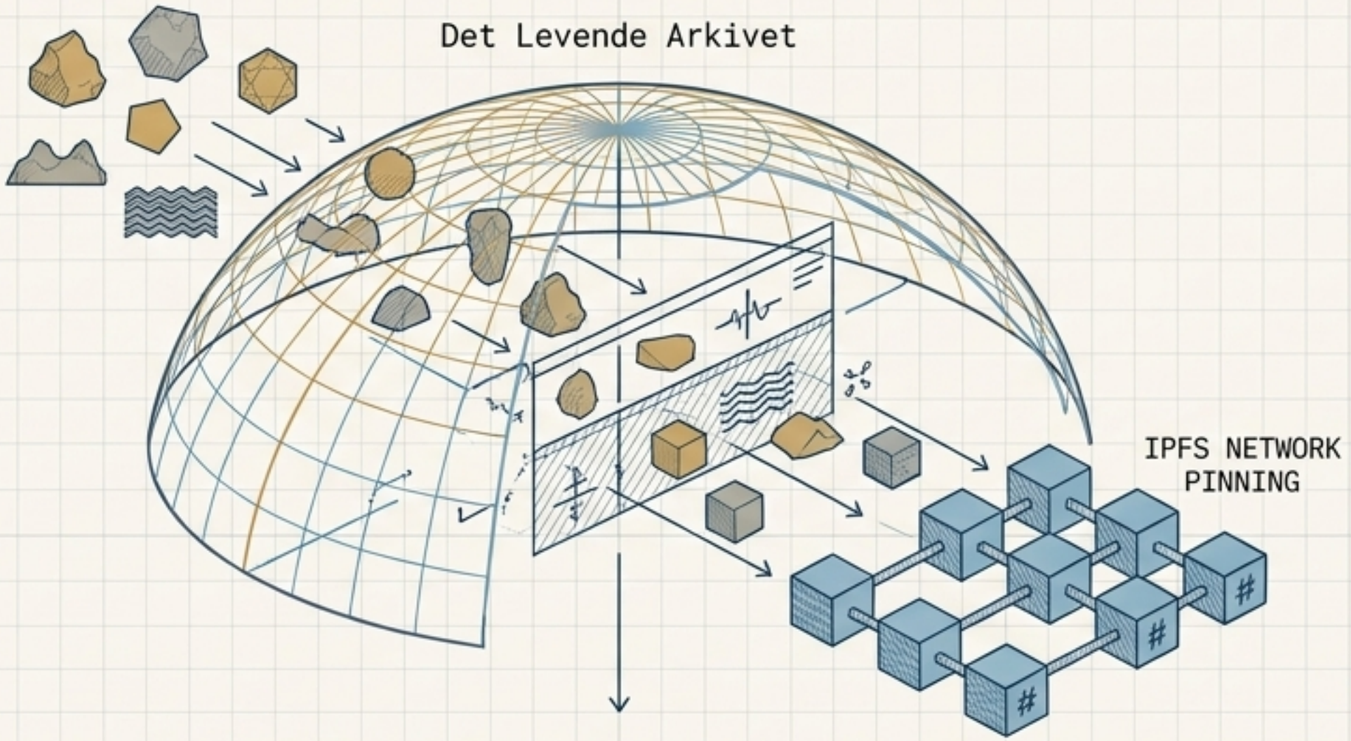
3. Ed25519 Signature Envelopes
The Composite Key is signed via asymmetric cryptography (RFC 8032), capturing the author's public key without without centralized PKI.

4. Hash-Linked Sequence
`seq 001` (`Genesis`) links to `seq 002` via the `prev` field. Altering any historical record invalidates every subsequent link.



Gemini Notebook

Det Levende Arkivet: The Autonomous Ingestion Engine



Manual file uploads introduce operational friction, human delay, and recording bias. Det Levende Arkivet (The Living Archive) operates as an automated multi-layer pipeline that captures physical real-world inputs, normalizes them, and pins them cryptographically to the IPFS network in real time.

Core Directives

1. Continuous, human-free data ingestion.

Core Directives

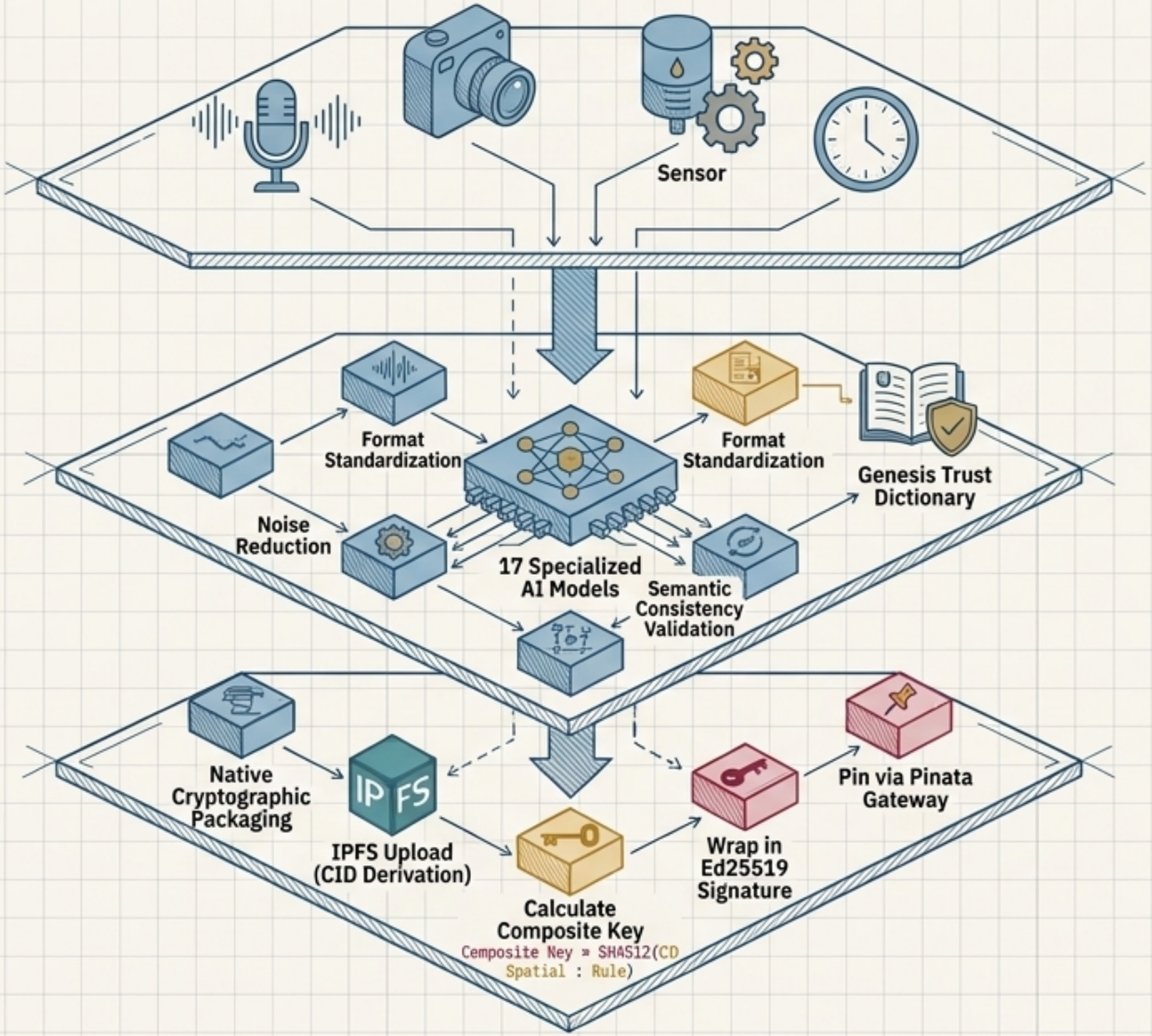
2. Real-time cryptographic validation.

Core Directives

3. Verifiable historical record-keeping.

Gemini Notebook

The 3-Layer Ingestion Pipeline



Layer 1: The Senses (Sansene / Physical Layer)

Automated real-world collection: Sensors, microphones, cameras, and cryptographic timestamps.

Output: Raw physical sensor inputs.

Layer 2: The Brain (Hjernen / AI Processing Layer)

An array of 17 specialized AI models executes noise reduction, format standardization, and semantic consistency validation against the Genesis Trust Dictionary.

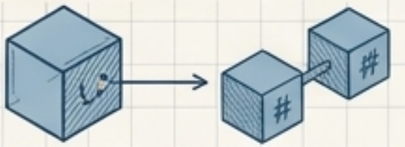
Output: Canonical JSON / Structured text payload.

```
{
  "canonical": true,
  "timestamp": "2024-05-15T10:30:00Z",
  "data": [...]
}
```

Layer 3: The Nerve (Nerven / CVP Protocol Layer)

Executes native cryptographic packaging. Uploads to IPFS (CID derivation), calculates Composite Key, wraps in Ed25519 signature, and pins via Pinata Gateway.

Output: Hash-linked .car archive.



Gemini Notebook

Native .car Format & The Insertion Protocol



The Container

CVP adopts the .car (**Content Addressable Record**) as a portable, chain-agnostic container standard. It holds the payload, vectors, keys, and signatures in one file.

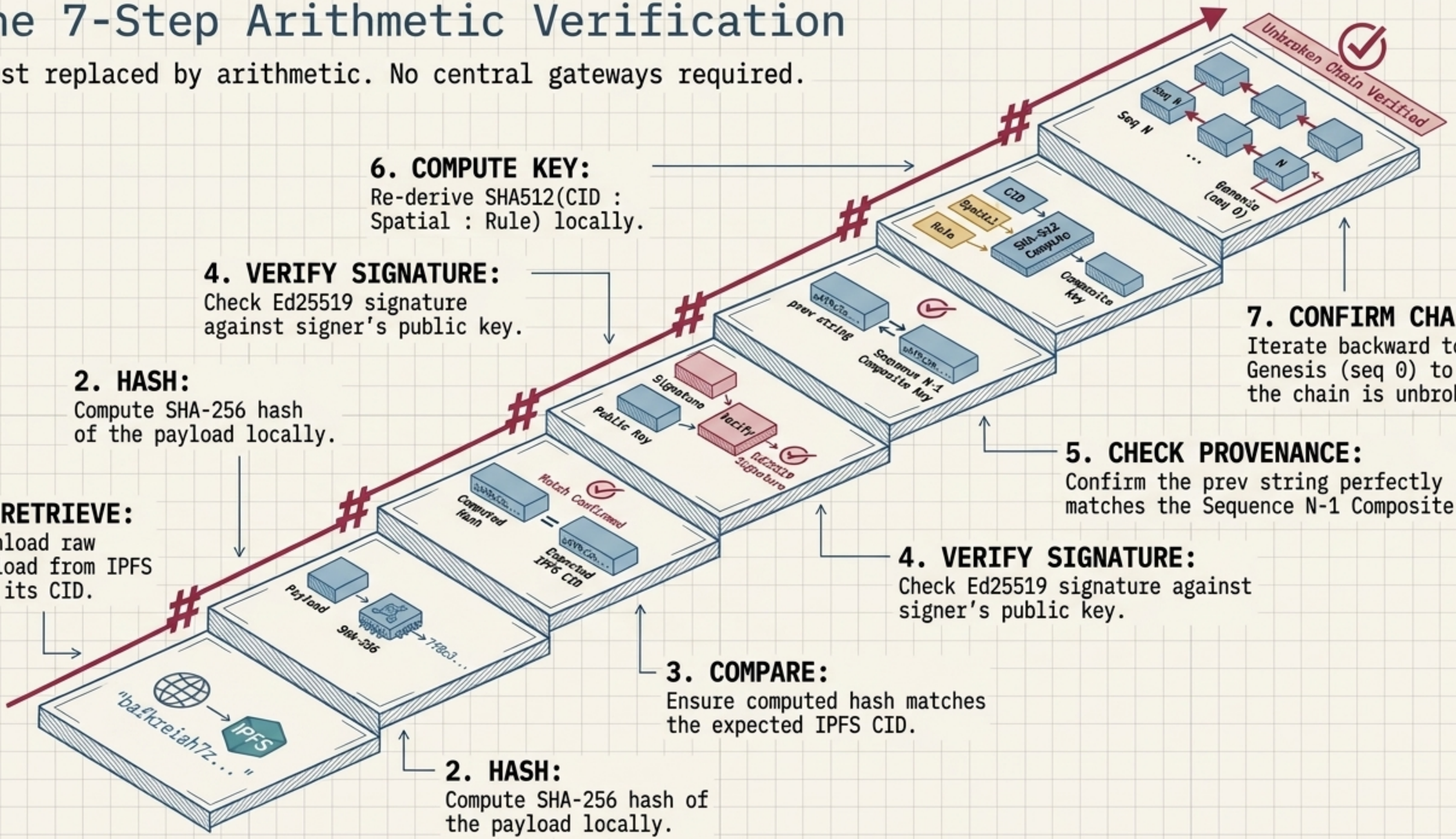
Insertion Protocol

Allows complex documents to reference external instruments (e.g., car://003-fingerprint) without duplicating payload data, incorporating verifiable states seamlessly.

Gemini Notebook

The 7-Step Arithmetic Verification

Trust replaced by arithmetic. No central gateways required.



1. RETRIEVE:
Download raw payload from IPFS via its CID.

2. HASH:
Compute SHA-256 hash of the payload locally.

3. COMPARE:
Ensure computed hash matches the expected IPFS CID.

4. VERIFY SIGNATURE:
Check Ed25519 signature against signer's public key.

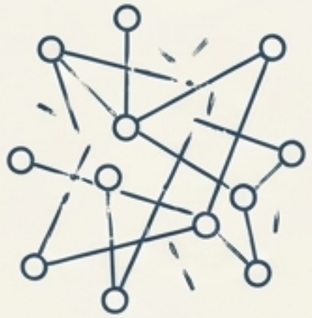
5. CHECK PROVENANCE:
Confirm the prev string perfectly matches the Sequence N-1 Composite Key.

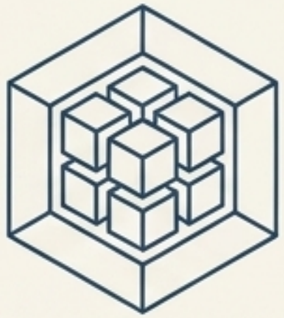
6. COMPUTE KEY:
Re-derive SHA512(CID : Spatial : Rule) locally.

7. CONFIRM CHAIN:
Iterate backward to Genesis (seq 0) to prove the chain is unbroken.

Gemini Notebook

Ecosystem Impact: EVM Smart Contracts vs. Native IPFS





EVM L1/L2 Provenance (CID stored in Smart Contract)	CVP Native IPFS Provenance (.car Envelopes)
Cost: High variable gas fees per transaction.	**Cost:** Zero Gas Costs (gasless execution).
Speed: High latency due to block confirmation times.	**Speed:** Sub-second network propagation.
Scale: Vulnerable to throughput bottlenecks.	**Scale:** Unlimited horizontal bandwidth.
Integrity: CIDs are passive pointers; metadata is disconnected off-chain.	**Integrity:** Self-contained vectors, rules, and signatures in a unified container.

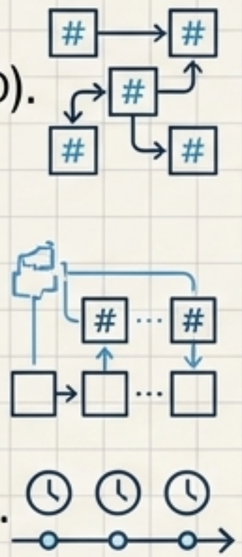
Gemini Notebook

Protocol Priorities vs. Legal Boundaries (§15 Non-Goals)

CVP provides cryptographic priority—unalterable factual evidence. It does not usurp statutory legal systems.

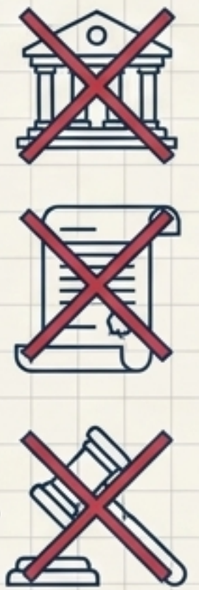
WHAT CVP PROVIDES
(Protocol Level)

- [✓] Proof of Unaltered Content (`CID`).
- [✓] **Proof of Signing Key Authenticity** (`Ed25519`).
- [✓] Immutable Sequential Ordering (`Hash Chain`).
- [✓] Verifiable **Prior Art** Timestamps.



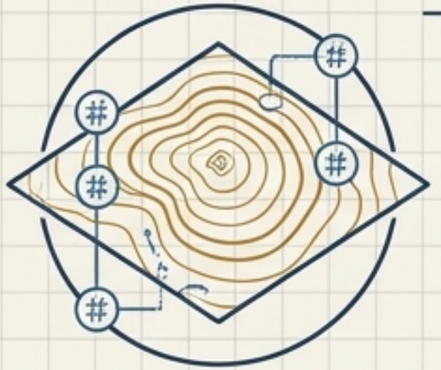
WHAT CVP DOES NOT CLAIM
(Legal Level)

- [✗] Override of **State/Statutory Jurisdiction.**
- [✗] Creation of **Statutory Legal Title.**
- [✗] Immunity from External Law.
- [✗] Automated Judicial Enforcement.



Gemini Notebook

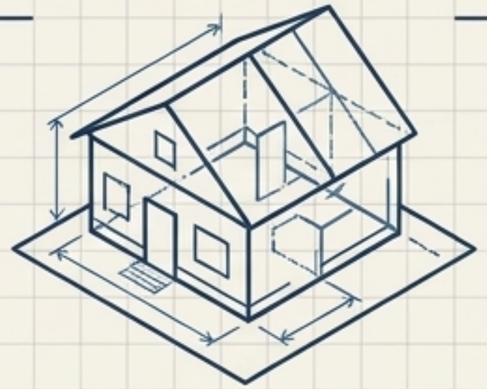
Real-World Implementation Scenarios



Scenario A: **Tribal Land Stewardship**

Protecting ancestral territories from forged registry filings.

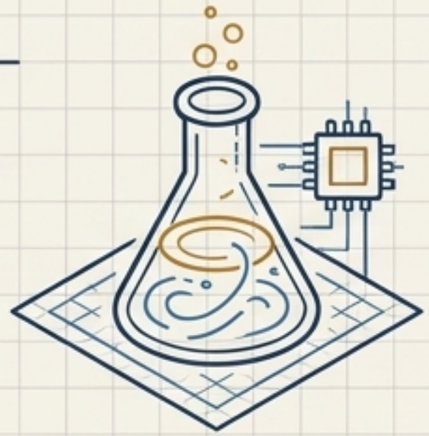
Georeferenced coordinates and oral histories mapped to CIDs, signed by a 3-of-5 Elders multi-sig key. Proves continuous occupancy.



Scenario B: **Private Custody Protection**

Permanent, private records for children's health and history.

Local .car.json objects signed with private keys. Composite keys mathematically prove documents existed unedited at exact timestamps.



Scenario C: **Scientific Prior Art**

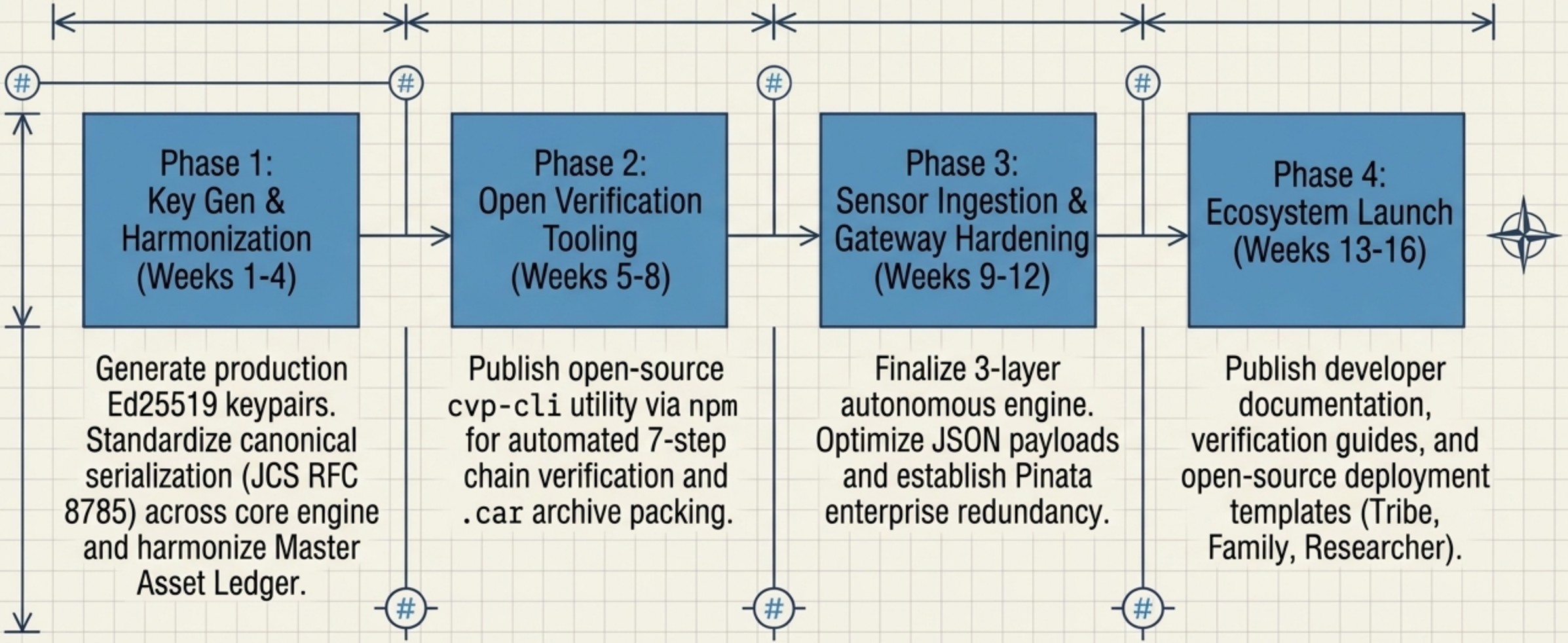
Establishing priority over intellectual property before journal publication.

Code and lab notes pinned to IPFS, wrapped in CVP rules. Creates undeniable, timestamped proof of priority.

Gemini Notebook

Project Roadmap & Technical Backlog

Resolving technical debt and building the open-source scaffold.



Gemini Notebook

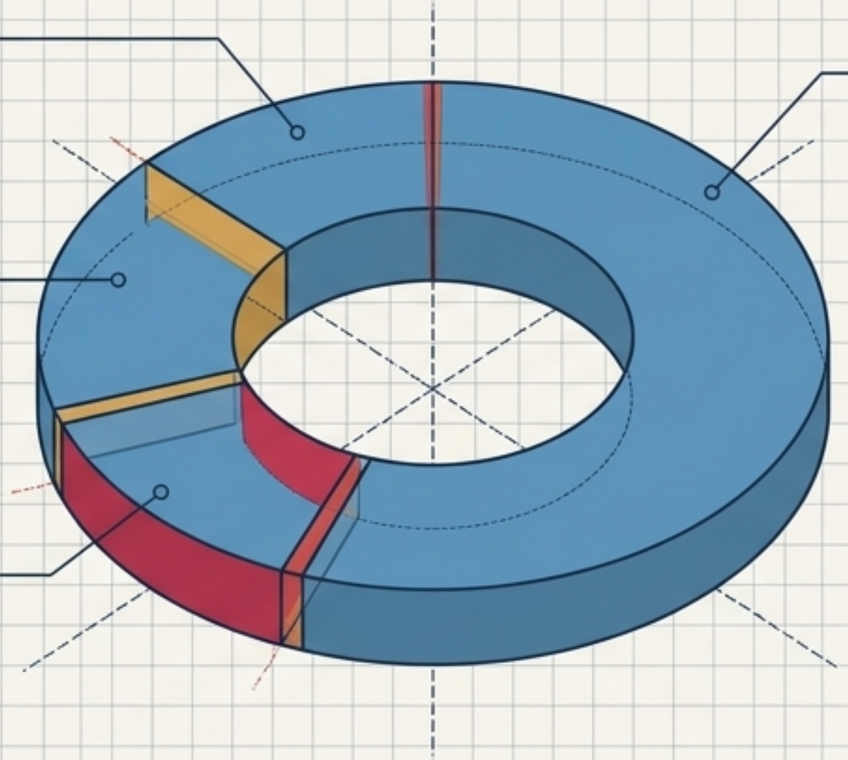
Grant Budget Allocation

Total Ask: $50,000 USD

$6,500 (13%)
IPFS Infrastructure & Storage
Dedicated nodes, Pinata enterprise pinning, long-term archival redundancy.

$8,500 (17%)
Documentation & Community Scaffold
Walkthroughs, legal boundary maps, and public verification guides.

$7,000 (14%)
Testing & Release Tooling
Security auditing, cross-platform validation (Linux/macOS/Windows), npm distribution.

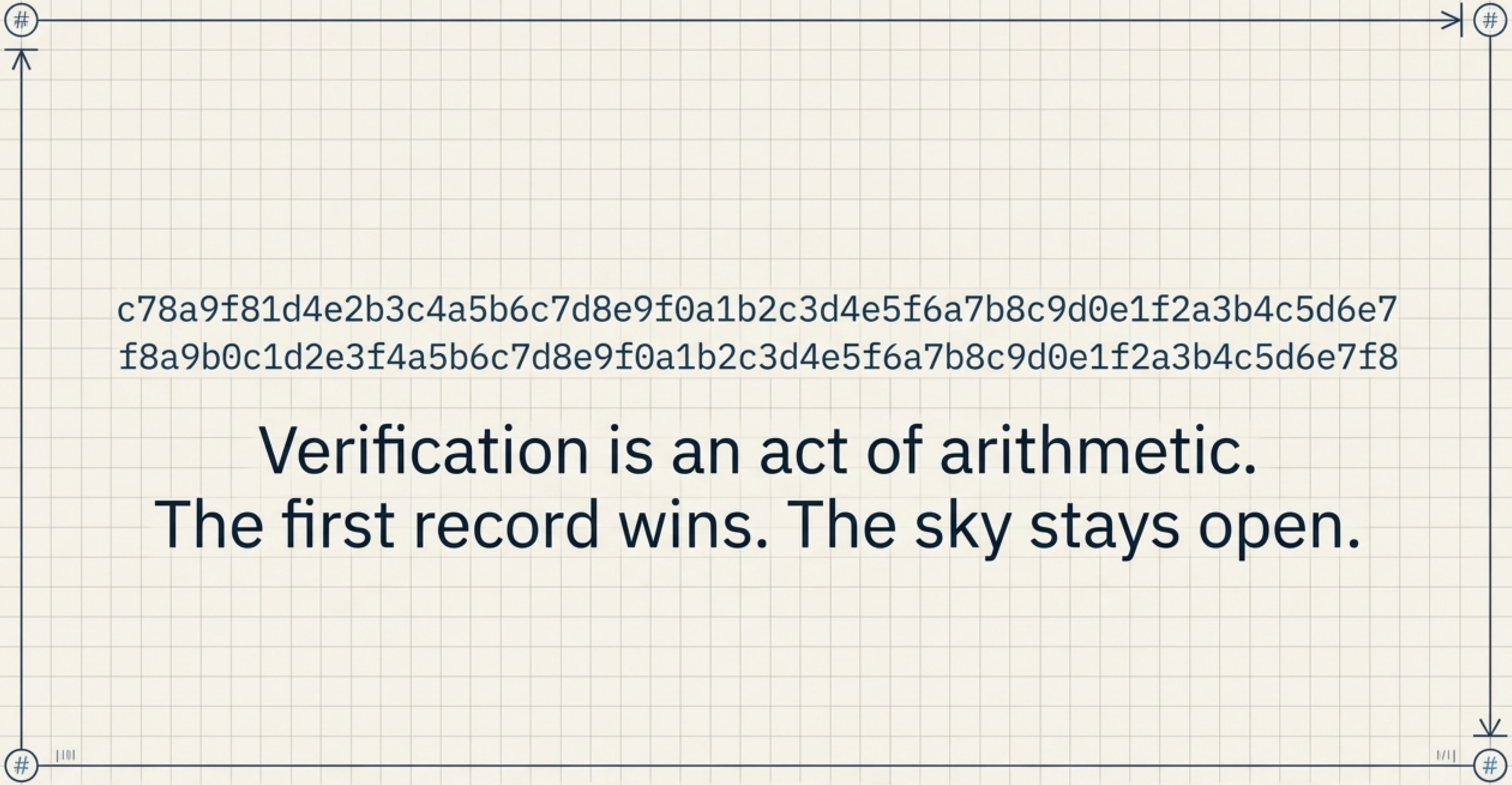


$28,000 (56%)
Cryptographic & Protocol Engineering

JCS RFC 8785 implementation, Ed25519 refactoring, cvp-cli development.

Demonstrating extreme capital efficiency to bring CVP from draft specification to a fully deployed IPFS toolset.

Gemini Notebook



c78a9f81d4e2b3c4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7
f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8

Verification is an act of arithmetic.
The first record wins. The sky stays open.

Trust Namespace: commons-vector-v1

Primary Infrastructure: IPFS & Pinata Archival Gateway

Submitted by: Kim Terje Rudschinat Grønli

Gemini Notebook